PROJECT CASE STUDY

Cricket Ball & Bat Tracking

Detection, Hit Events, Trajectory & Speed

Category	Status	Year
Computer Vision	Completed prototype	2025

A YOLOv8-based vision pipeline for ball and bat detection, motion tracking, hit-event reasoning, trajectory reconstruction, and real-time speed estimation.

YOLOv8n	3D	Real-time
detector	trajectory model	tracking goal

Context and problem

A cricket ball is small, fast, frequently blurred, and often occluded, making reliable tracking and speed estimation difficult from ordinary video.

Strategy and implementation

- Fine-tuned YOLOv8n for ball and bat bounding boxes.
- Used tracking plus bounding-box interaction to identify hit events.
- Applied optical-flow fallback for difficult frames.
- Implemented Tomasi-Kanade-based 3D trajectory reconstruction for speed estimation.

My role

Dataset preparation, YOLO training, event logic, tracking integration, trajectory work, and application-level visualization.

Technology stack

YOLOv8 | Ultralytics | OpenCV | Python | Optical Flow | Tomasi-Kanade

Results and evidence

- Produced ball-bat detections and tracked motion through video.
- Connected detection output to hit-event and speed logic.
- Built a practical computer-vision system rather than a standalone model notebook.

Key takeaways

- Small-object detection needs temporal reasoning after the detector.
- A useful sports system combines detection, tracking, geometry, and UI feedback.

Portfolio: github.com/awai1s | awaisraza5424@gmail.com